Cagatay Ayhan

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PROFILE

Currently seeking postdoctoral and industry research opportunities beginning in Summer or Fall 2027. I am a Ph.D. candidate in Biomathematics at Florida State University, advised by Tom Needham. My research develops and applies methods from topological data analysis, geometry, and machine learning to scientific data, including neuroscience and materials science. I welcome inquiries about postdoctoral research and industry roles in data science, applied science, and related quantitative fields.

EDUCATION

Ph.D. Candidate & M.S. in Biomathematics August 2021 – May 2027
Florida State University, Tallahassee, FL
M.S. conferred Fall 2025 | GPA: 3.67/4.0

B.S. in Pure Mathematics July 2015 – June 2021
Ege University, Izmir, Turkey
GPA: 3.70/4.0, High Honors | Exchange student in mathematics, University of Aberdeen, Scotland (Fall 2017 and Fall 2018)

RESEARCH & PROFESSIONAL EXPERIENCE

Georgia Tech Research Institute Summer 2026
Research Intern, Atlanta, GA
Explored applications of topological data analysis to materials science under the supervision of Dr. Branden Stone. The internship has concluded, and the research collaboration is ongoing.

Florida State University, Department of Mathematics Fall 2021 – Present
Tallahassee, FL

- **Research Assistant:** NSF Grant DMS-2324962 (Spring 2026; Summer 2025; Fall and Summer 2024)
- **Course Instructor:** MAP2302 Ordinary Differential Equations (Spring 2027); MAS3105 Applied Linear Algebra (Fall 2026, Summer 2026); MAC2313 Calculus III (Fall 2025); MAC2312 Calculus II (Spring 2025); MAC2311 Calculus I (Spring 2024); MAC1140 Precalculus Algebra (Spring 2023)
- **Additional Teaching:** Recitation instructor for MAD2104 Discrete Mathematics I (Fall 2023) and MAC2311 Calculus I (Fall 2022); lab/lecture assistant for MAC2233 Business Calculus (Summer 2023, Fall 2021); teaching assistant for MGF1107, MAC1140, and MAC1105 (Summer and Spring 2022); grader for MAC3301 and MAC2302 (Spring 2022, Fall 2021)

PUBLICATIONS

Published

- **Cagatay Ayhan** and Tom Needham (2026). *Equivalence of Landscape and Erosion Distances for Persistence Diagrams*. *Advances in Applied Mathematics*. Journal version | arXiv:2506.21488.

Accepted

- **Cagatay Ayhan**, Audrey N. Nash, Roberto Vincis, Martin Bauer, Richard Bertram, and Tom Needham (2025). *A Persistent Homology Pipeline for the Analysis of Neural Spike Train Data*. Accepted for publication in *SIAM Journal on Life Sciences*; to appear. arXiv:2512.08637.

TECHNICAL SKILLS

Programming and data: Python, Pandas, NumPy, scikit-learn, Ripser, GUDHI | **Tools:** LaTeX, Microsoft Office
Languages: Turkish (native), English (C2 professional proficiency)

PRESENTATIONS

Sept. 23, 2026 *Equivalence of Landscape and Erosion Distances for Persistence Diagrams*. Invited talk at the Math and Data Seminar, Rutgers University, Piscataway, NJ.

March 2026 *A Persistent Homology Pipeline for the Analysis of Neural Spike Train Data*. AMS 2026 Spring Southeastern Sectional Meeting, Georgia Southern University, Armstrong Campus, Savannah, GA.

January 2026 *Equivalence of Landscape and Erosion Distances for Persistence Diagrams*. Talk at the UF-FSU Topology and Geometry Meeting, University of Florida, Gainesville, FL.

November 2025 *Equivalence of Landscape and Erosion Distances for Persistence Diagrams*. Talk and poster at the Topological Data Analysis Conference 2025, University of Missouri, Columbia, MO.

October 2025 *Equivalence of Landscape and Erosion Distances for Persistence Diagrams*. Online talk at ComPer 2025: Workshop on Computational Topology, University at Albany, State University of New York.

September 2025 *Equivalence of Landscape and Erosion Distances for Persistence Diagrams*. FSU Data Science Seminar, Florida State University, Tallahassee, FL.

August 2025	*A Topological Framework for Network Activity*. Poster at the NITMB Math-Bio Convergence Conference, Chicago, IL.
April 2025	*A Topological Framework for Simultaneously Recorded Network Activity*. Graduate Research Week Flash Talk Competition, Florida State University, Tallahassee, FL.
March 2025	*Topological Optimization Techniques*. Machine Learning/Data Science Seminar, Florida State University, Tallahassee, FL.
March 2024	*TDA in Simultaneously Recorded Neural Spike Train Datasets*. AMS 2024 Spring Southeastern Sectional Meeting, Florida State University, Tallahassee, FL.
November 2023	*Methods in Topological Data Analysis with an Application in Simultaneously Recorded Neural Spike Train Datasets*. ATE candidacy exam, Florida State University, Tallahassee, FL.
October 2023	*Multiple Neuron Spike Train Data Analysis Using Persistent Homology*. Lightning talk at the ICERM Topology and Geometry in Neuroscience Workshop, Brown University, Providence, RI.

HONORS & TRAINING

Awards and Honors

- Tam Family Award for Excellence in Teaching and Research
- Dewitt Sumners Flash Talks Competition, First Place (tied), April 2025
- Bettye Anne Bushee Case Graduate Fellowship, April 2024
- Outstanding Teaching Assistant Award, nominated twice, 2023–2024

Data Science Training

August 2025	Data Science Boot Camp, The Erdős Institute. GitHub project: Decoding Human Activity.
May 2024	Data Science Boot Camp, The Erdős Institute. GitHub project: Tallahassee Crime Map.
December 2023	Data Science Boot Camp, The Erdős Institute. GitHub project: Near Earth Objects.

CONFERENCES & WORKSHOPS

Sep. 14–18, 2026	Connectomics, Non-Euclidean data, and Statistics: Tutorial and Hands-on, Institute for Mathematical and Statistical Innovation (IMSI), University of Chicago, Chicago, IL.
Jul. 27–29, 2026	Graph Based Methods in Topological Data Analysis, Michigan State University, STEM Building, East Lansing, MI.
March 2026	AMS 2026 Spring Southeastern Sectional Meeting, Georgia Southern University, Armstrong Campus, Savannah, GA.
January 2026	UF-FSU Topology and Geometry Meeting, University of Florida, Gainesville, FL.
November 2025	Conference on Topological Data Analysis: Recent Developments and Applications, Columbia, MO.
August 2025	NITMB Math-Bio Convergence Conference, Chicago, IL.
June 2024	AMS Mathematics Research Communities: Climate Science at the Interface Between Topological Data Analysis and Dynamical Systems Theory, Buffalo, NY.
March 2024	AMS 2024 Spring Southeastern Sectional Meeting, Florida State University, Tallahassee, FL.
November 2023	UF-FSU Topology Meeting, Florida State University, Tallahassee, FL.
October 2023	Topology and Geometry in Neuroscience Semester Workshop, ICERM, Brown University, Providence, RI.

PROFESSIONAL SERVICE

Co-organizer, with Tom Needham and Martin Bauer, Special Session on *Geometric Methods for Data Science*, 2026 Spring Southeastern Sectional Meeting of the American Mathematical Society, Georgia Southern University, Armstrong Campus, Savannah, GA, March 28–29, 2026.

OUTREACH

FSU Math Fun Day Volunteer

Annual public outreach event hosted by the FSU Department of Mathematics.

- Guided hands-on activities on fractal geometry in the Creating Geometric Patterns room (February 1, 2025).
- Assisted students with puzzles at the Geometric Constructions station (March 25, 2023).

Invited Online Talk for Undergraduate Students May 29, 2023

Ege University, Turkey

Spoke about pursuing graduate study in the United States and shared practical guidance on the application process.

Last updated: October 2026